GEOMETRY QUEST · WORLD 8

3D Shapes & Nets

ANSWER SHEET · for the grown-up

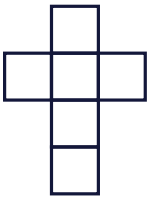
Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

1 Complete the table. Then check that faces + vertices − edges = 2 for every shape.

shape	faces	edges	vertices	F + V − E
cube	**6**	**12**	**8**	**2**
cuboid	**6**	**12**	**8**	**2**
triangular prism	**5**	**9**	**6**	**2**
square-based pyramid	**5**	**8**	**5**	**2**
tetrahedron	**4**	**6**	**4**	**2**

2 Which of these nets fold up into a cube? Tick them. (Imagine folding, or copy them onto squared paper and try.)

A

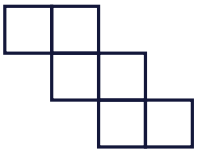


yes

B

no

C



yes

D

no

3 Name the 3D shape.

I have 6 square faces. — **cube**

1 square face + 4 triangles meeting at a point. — **square-based pyramid**

I have 2 flat circles and 1 curved surface. — **cylinder**

I have one curved surface and no edges at all. — **sphere**

4 Cut out the dice net on the next page, fold along the lines and glue the tabs. Opposite faces of a real dice add up to 7. Do yours?

Yes: 1↔6, 2↔5, 3↔4 end up opposite.

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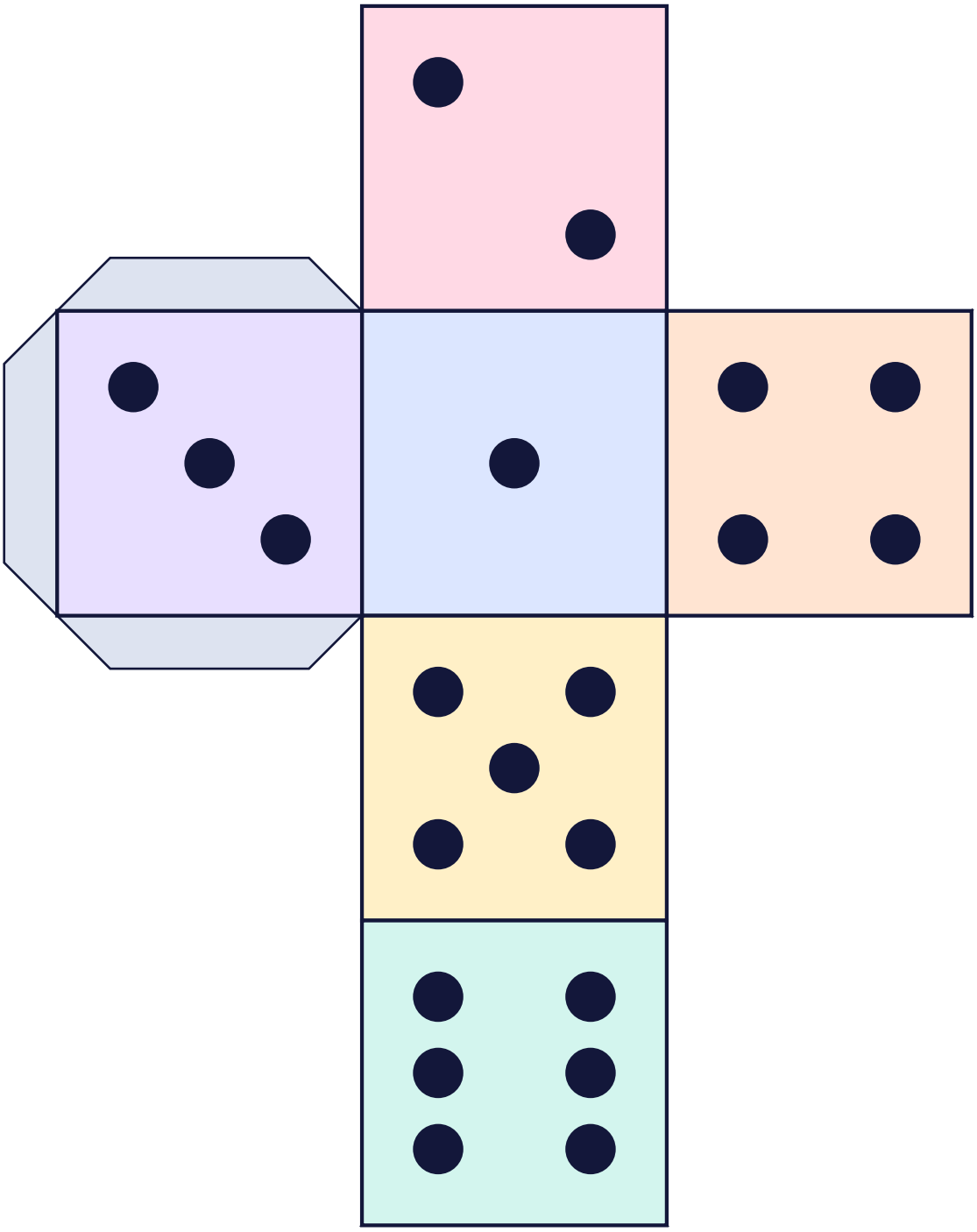
Cut out, fold along every line, glue the grey tabs. Build your own dice!



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